2. To Create Main Loop, Display Prompt, Read User Input, Handle Exit Conditions, Design Control Flow Diagram, Test Interactive Loop

Code:

```
  GNU nano 8.7.1
#include <stdio.h>
#include <string.h>

#define MAX_INPUT 100

int main() {
    char input[MAX_INPUT];

    printf("Simple Interactive Shell\n");
    printf("Type 'exit' to quit.\n\n");

    // Main interactive loop
    while (1) {

        // Display prompt
        printf("shell> ");

        // Read user input
        if (fgets(input, sizeof(input), stdin) == NULL) {
            printf("\nInput error or EOF detected.\n");
            break;
        }

        // Remove newline character
        input[strcspn(input, "\n")] = '\0';

        // Handle empty input
        if (strlen(input) == 0) {
            continue;
        }

        // Handle exit condition
        if (strcmp(input, "exit") == 0) {
            printf("Exiting shell...\n");
            break;
        }

        // Display entered command
        printf("Command entered: %s\n", input);
    }

    printf("Shell terminated successfully.\n");

    return 0;
}
```

Result:

```
bheeshma@SINGAM:~/SKILL$ nano Skill2a.c
bheeshma@SINGAM:~/SKILL$ gcc Skill2a.c -o S2a
bheeshma@SINGAM:~/SKILL$ ./S2a
Simple Interactive Shell
Type 'exit' to quit.

shell> hello
Command entered: hello
shell> ls
Command entered: ls
shell> pwd
Command entered: pwd
shell> test
Command entered: test
shell> exit
Exiting shell...
Shell terminated successfully.
bheeshma@SINGAM:~/SKILL$
```

To Capture Keyboard Input, Handle Backspace, Process Enter Key, Manage Input Buffer, Support Multi-Character Commands, Test User Interaction

Code:

```
  GNU nano 8.7.1
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <termios.h>

#define BUFFER_SIZE 100

// Enable character-by-character input
void enableRawMode(struct termios *oldSettings) {
    struct termios newSettings;

    tcgetattr(STDIN_FILENO, oldSettings);
    newSettings = *oldSettings;

    newSettings.c_lflag &= ~(ICANON | ECHO);

    tcsetattr(STDIN_FILENO, TCSANOW, &newSettings);
}

// Restore normal terminal input
void disableRawMode(struct termios *oldSettings) {
    tcsetattr(STDIN_FILENO, TCSANOW, oldSettings);
}

int main() {

    char inputBuffer[BUFFER_SIZE];
    int position = 0;
    char ch;

    struct termios oldSettings;

    printf("Interactive Keyboard Input\n");
    printf("Type a command and press Enter.\n");
    printf("Use Backspace to delete characters.\n");
    printf("Type 'exit' to quit.\n\n");

    enableRawMode(&oldSettings);

    while (1) {

        printf("shell> ");
        fflush(stdout);

        position = 0;
        memset(inputBuffer, 0, BUFFER_SIZE);

        while (1) {

            // Capture keyboard input
            if (read(STDIN_FILENO, &ch, 1) <= 0) {
```

```
  GNU nano 8.7.1
            // Capture keyboard input
            if (read(STDIN_FILENO, &ch, 1) <= 0) {
                break;
            }

            // Handle Enter key
            if (ch == '\n' || ch == '\r') {

                inputBuffer[position] = '\0';

                printf("\n");

                // Exit condition
                if (strcmp(inputBuffer, "exit") == 0) {
                    disableRawMode(&oldSettings);

                    printf("Exiting shell...\n");
                    return 0;
                }

                // Display entered command
                if (position > 0) {
                    printf("Command entered: %s\n", inputBuffer);
                }

                break;
            }

            // Handle Backspace
            else if (ch == 127 || ch == 8) {

                if (position > 0) {

                    position--;

                    inputBuffer[position] = '\0';

                    // Erase character from terminal
                    printf("\b \b");
                    fflush(stdout);
                }
            }

            // Store normal characters
            else if (position < BUFFER_SIZE - 1) {

                inputBuffer[position] = ch;
                position++;

                // Display typed character
                printf("%c", ch);
```

```
                // Display typed character
                printf("%c", ch);
                fflush(stdout);
            }
        }
    }

    disableRawMode(&oldSettings);

    return 0;
}
```

Result:

```
bheeshma@SINGAM:~/SKILL$ nano Skill2b.c
bheeshma@SINGAM:~/SKILL$ gcc Skill2b.c -o S2b
bheeshma@SINGAM:~/SKILL$ ./S2b
Interactive Keyboard Input
Type a command and press Enter.
Use Backspace to delete characters.
Type 'exit' to quit.

shell> hello
Command entered: hello
shell> ls -l
Command entered: ls -l
shell> echo hello world
Command entered: echo hello world
shell> helo
Command entered: helo
shell> exit
Exiting shell...
bheeshma@SINGAM:~/SKILL$
```